Literature-implied answer for arXiv:2009.05503

The logarithmic-degree level inequality follows from sharp hypercontractivity

Status. literature_implied_answer. This packet records a quantitative identification with a later theorem; it does not claim a new proof of the sharp hypercontractive inequality.

Source question. In Section 7.1, "Global functions are concentrated on the high degrees" (PDF page 30), Filmus–Kindler–Lifshitz–Minzer write, immediately after restating their Theorem 1.6:

> It would be interesting to prove a quantitatively better version of Theorem 1.6 in terms of d, and in particular whether it is the case that for $d = c\log(1/\varepsilon)$ it holds that $\|f^{=d}\|_2^2 = \varepsilon^{2-o(1)}$ for sufficiently small (but constant) $c > 0$.

The surrounding paragraph concerns an improved upper bound. Literal equality cannot be intended for every admissible function (take $f = 0$), so below the display is read as the requested upper-bound scale.
Here a Boolean function $f : S_n \to \{0, 1\}$ is $(2d, \varepsilon)$-global if every consistent restriction fixing at most $2d$ input-output pairs has L^2-norm at most ε.

Later theorem. Keevash–Lifshitz, arXiv:2307.15030, Theorem 4.1 (PDF page 20), proves the following. If $g \in L^2(S_n)$ is $(r, \gamma_1, \gamma_2, d)$-biglobal, $r > 1$, and

$$d \le \min\left\{\frac{1}{4}\log(\gamma_2/\gamma_1), 10^{-5}n\right\},$$

then

$$\|g^{=d}\|_2^2 \le \gamma_1^2\left(10^6 r^2 d^{-1}\log(\gamma_2/\gamma_1)\right)^d.$$

Their Definition 2.4 says that biglobalness means every restriction of size $t \le d$ has L^1-norm at most $r^t\gamma_1$ and L^2-norm at most $r^t\gamma_2$.

Parameter identification. Assume $0 < \varepsilon < 1$ and that $f : S_n \to \{0, 1\}$ is $(2d, \varepsilon)$-global in the 2020 sense. For every restriction f_* of size at most d,

$$\|f_*\|_2 \le \varepsilon, \qquad \|f_*\|_1 = \|f_*\|_2^2 \le \varepsilon^2,$$

where the identity uses that f_* remains Boolean. Thus f is $(2, \varepsilon^2, \varepsilon, d)$-biglobal. Theorem 4.1 gives

$$\|f^{=d}\|_2^2 \le \varepsilon^4\left(4\cdot 10^6 d^{-1}\log(1/\varepsilon)\right)^d \tag{1}$$

provided

$$d \le \min\left\{\frac{1}{4}\log(1/\varepsilon), 10^{-5}n\right\}.$$

Set $L = \log(1/\varepsilon)$ and $d = cL$ (rounding to an integer if needed). Then the right-hand side of (1) is

$$\varepsilon^{4-c\log(4\cdot 10^6/c)}.$$

As $c \downarrow 0$, the exponent tends to 4. In particular, one may choose an absolute $c > 0$ sufficiently small that the exponent is strictly larger than 2 (indeed, larger than 3). This is stronger than an $\varepsilon^{2-o(1)}$ upper bound.

Conclusion. *The intended logarithmic-degree upper-bound question in arXiv:2009.05503 has an affirmative answer by Keevash–Lifshitz, Theorem 4.1. Their theorem also replaces the source's very large lower bound on n by the essentially linear condition $d \le 10^{-5}n$ for this application.*

Scope. This identification answers the explicit quantitative level-d subquestion. It does not claim that every separate technical condition in all of the 2020 paper's hypercontractive theorems can be removed, nor does it address the broader junta-method problem for arbitrary product-free subsets of A_n. The Kruskal–Katona stability question later mentioned in the 2020 paper is answered in that same paper and is therefore an extraction false positive.

Search evidence. The run's cheap indexes contained no packet for arXiv:2009.05503. Searches for the paper title, "level-d", "global", and "symmetric group hypercontractivity" led to arXiv:2307.15030. Its source was checked at Definition 2.4 and Theorem 4.1, and the parameter substitution above was performed directly.

References

[1] Y. Filmus, G. Kindler, N. Lifshitz, and D. Minzer, *Hypercontractivity on the symmetric group*, arXiv:2009.05503 (2020).

[2] P. Keevash and N. Lifshitz, *Sharp hypercontractivity for symmetric groups and its applications*, arXiv:2307.15030 (2023).